G0251 FOUNDATION OF BUSINESS ANALYTICS
TUTORIAL 3

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Question 1

The table below contains information about people's ages and weights.

Name	Age	Weight
Alice	25	60
Bob	NaN	70
Carol	30	NaN
Dan	28	75

Based on this table, identify:

a. Which columns have missing values?

b. How many rows have at least one missing value?

c. Suggest one way to fill in the missing Age and Weight values.

Question 2

The table below contains information about the heights of five individuals.

Person	Height (cm)
John	170
Sarah	165
Alex	180
Emily	172
Kevin	250

Based on this table, identify:

a. Which height might be considered unusual at first glance?

b. If a height is considered an outlier when it is greater than 200 cm or less than 100 cm, how many outliers are there?

c. Suggest what can be done to handle the unusual height.

Question 3

The table below shows the air quality, measured by a device.

Time	Air Quality
8:00 AM	55
9:00 AM	60
10:00 AM	1200
11:00 AM	58

Based on the result of the air quality shown in the above table:

a. At what time does the device record an unusual Air Quality value?

b. Assuming normal values range between 0 and 100, how can the noisy data point be corrected?

c. What could cause such noisy readings?

Question 4

The table below shows the results of mid-term test of a group of high school students.

Student_ID	Name	Score
1	Alice	85
2	Bob	90
3	Alice	85
4	Carol	92

Based on the results displayed in the above table:

a. Are there any rows in the dataset that are repeated?

b. How many duplicate records are present?

c. Suggest how to remove the duplicate without losing any important information.

Question 5

The table below shows the records of sales by a fruit store. However, the data stored in the table has some errors.

Transaction_ID	Item	Price	Quantity
1	Apple	3	5
2	Banana	NaN	2
3	Orange	-1	10
4	Apple	3	5

Based on this table:

a. Which columns have missing values or unusual values?

b. Identify any duplicate transactions.

c. What price or quantity values seem incorrect, and how might they be fixed?

Question 6

A small company keeps records of employee hours and salaries but some errors have occurred in the data entry.

Employee_ID	Name	Hours_Worked	Salary
1	Alice	40	5000
2	Bob	NaN	6000
3	Carol	-5	4500
4	Alice	40	5000

Based on this table:

a. Identify the rows where:
   - There are missing values.
   - There are incorrect or unusual values (e.g., negative values).

b. How many duplicate records are present in the dataset?

c. Suggest how the following issues could be addressed:
   - The missing Hours_Worked for Bob.
   - The negative value for Carol's Hours_Worked.
   - Duplicate records.

d. Why is it important to fix such issues before analyzing the data?